



Primary school students during Ellicott STEM Night in Colorado, USA

challenges include a limited investor base and a small domestic market for robotics in India. Cost of adopting robotic technology is very high due to the cost of procuring imported hardware components as well as training personnel.”

Market growth catalysts

As the demand for STEM (science, technology, engineering and mathematics) skills in the job sector increases, there is a need for introducing these to coming generations with educational programmes and DIY activities right from school days. Several companies are coming up with educational toys to make learning fun. Parents and educators must come together to ensure its success. Dr Pravin Raut, founder and managing director, SINCOM Sindhu Electronics and Communications Pvt Ltd, says, “Looking at the role of electronic gadgets in our day-to-day life, it is now becoming necessary to learn the gems of electronics from school days, instead of college. With this motto, SINCOM-Training Division conducts various modules of Electronics Workshop to provide hands-on training, especially for school students. The objective of this workshop is to develop students’ interest in the electronics field so that they acquire basic skills in electronic devices from childhood and become smart kids.”

Atul Dumbre, director, Macfos Pvt Ltd, confirms, “STEM earlier used to be an optional subject, but now it is an integral part of education. Along with that, there are many coaching institutes

that are providing weekend or holiday training sessions with kits. This has created a buzz around kits.”

Vira elaborates on the need for experimental and experiential learning, “STEM learning wants people to understand that it is not just about new technologies, inventions or new medicines. Present-day needs of developing India require it to adopt several scientific and technological innovations. As a result, the education sector is facing a lot of challenges in meeting the skill demands. Offering a solution to this, STEM is emerging as the new alternative of futuristic education.”

Experts believe that government initiatives such as Atal Innovation Mission and Atal Tinkering Labs have boosted market growth. Hemant Parab, proprietor, Mega Kit, suggests, “The government and educational departments should think about revising the policies to incorporate more and more practicals along with theory in the syllabus to create the interest of students in the subject.”

These changes will make future generations smarter. Dileep Jain, chief executive officer, Rajguru Electronics (I) Pvt Ltd, explains, “Hands-on training to kids plays a major role, as nowadays many kids after working at school want to try at home. So, demand is created for the same.”

But what about college students who are forced to switch their domain when they lack the requisite skills? Parimal Wagh, senior administration officer, C-DAC, suggests partnerships between industry and academia, “The motivation of the faculty and profes-

sors to undertake projects in collaboration with the industry and experiment with kits for practical works act as major drivers for the growth in the education kits and technology.”

As a beginner, one often loses interest when stuck with a problem. Sometimes, even after a lot of research online, it gets difficult to determine the root cause. This is where the role of maker spaces and training services comes into play. Arun Barman, owner, Bormonz Technologies, says, “Good quality hands-on training programmes based on a student’s level of understanding should be implemented. It is true that theory without practical is empty, and practical without theory is blind. Many schools in cities have introduced STEM labs for learning with practicals, and almost all colleges organise hands-on training programmes conducted by training service providers. Thus, all manufacturers are looking for own or third-party training service providers so they can sell the training products/educational kits by providing training.”

Regarding the impact, Sagar Gala, director, Gala Electronics, conveys, “A robust STEM education creates critical thinkers, problem-solvers and next-generation innovators. Growth of STEM has picked up significantly over the last few years. India needs combined support from the government and other education societies to avail the opportunity and benefits of STEM education.”

Market challenges

For proper implementation of upgraded kits in schools and colleges throughout India, indigenous dealers still need to tackle a number of hurdles. In our economy, pricing plays an important role when it comes to competition among manufacturers. Jain says, “Mostly, the kits are a mixture of many IC components and modules. So, when an Indian company tries to make these kits, cost is higher because of the labour required to make them. International players like China get it cheaper and sometimes they make these kits automatically.”

About the present situation, Wagh says that the market is volatile, and it has proven that cost-effectiveness is very important. But customisation for